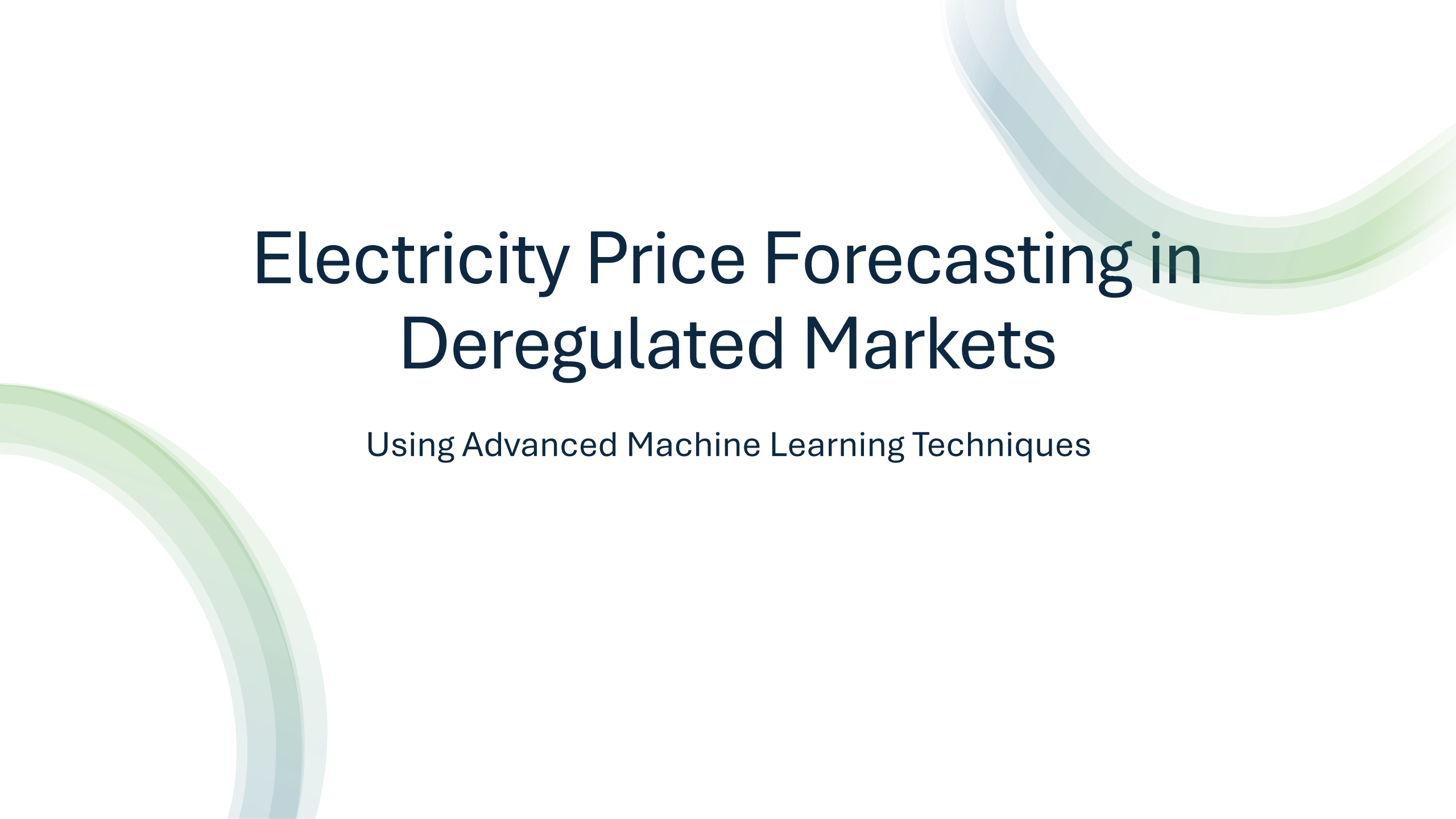




Electricity Price Forecasting in Deregulated Markets

Using Advanced Machine Learning Techniques

The Challenge of Electricity Price Forecasting

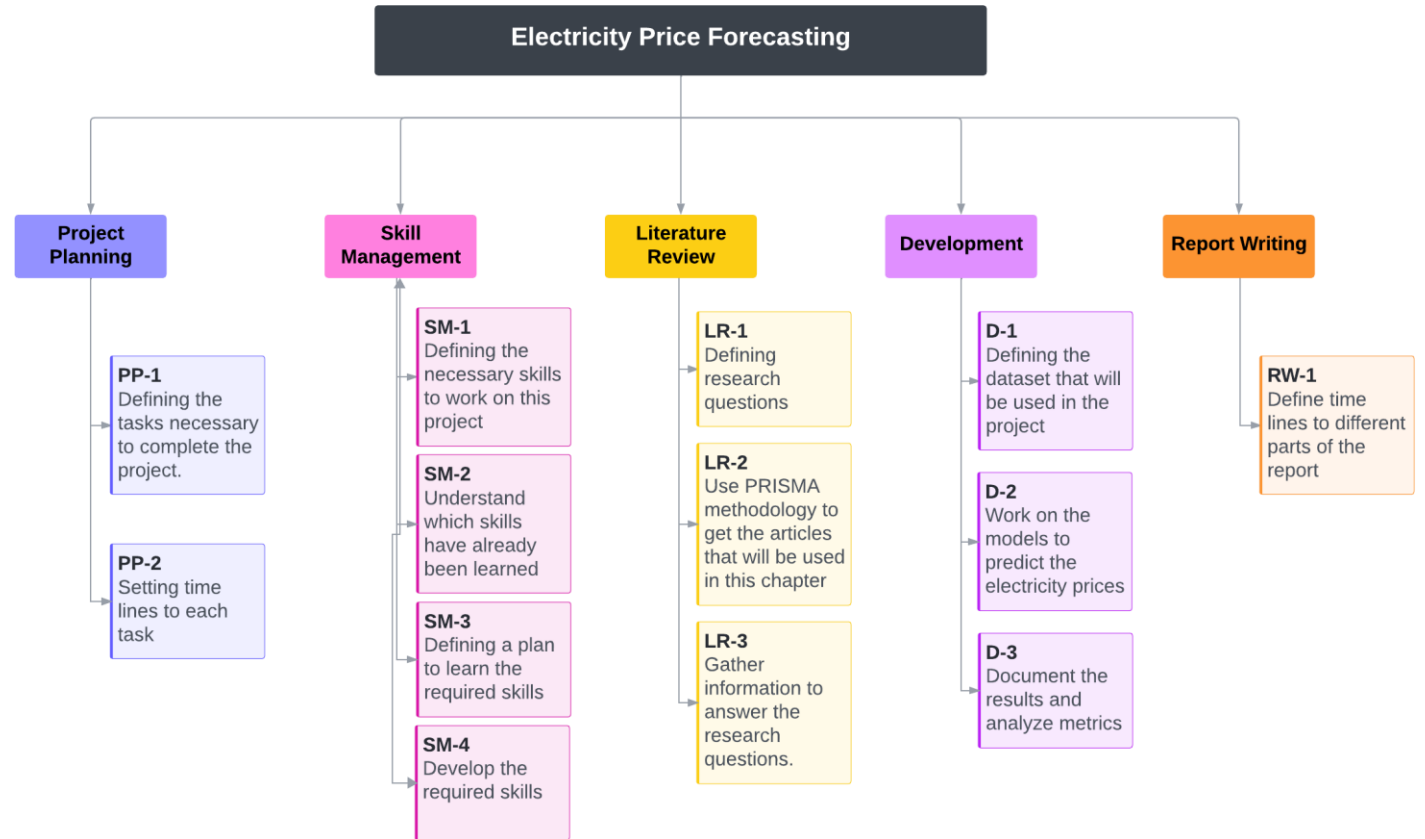
- Deregulated markets introduce price volatility due to:
 - Weather patterns (e.g., wind and solar variability)
 - Demand fluctuations
 - Supply-side uncertainties
 - Regulatory changes
- Traditional models struggle with non-linear, complex dynamics



Objectives

- Develop accurate machine learning models for:
 - Short-term forecasting: Predict prices hourly or daily.
 - Long-term forecasting: Predict prices monthly or yearly.
- Evaluate machine learning algorithms: LSTM, Random Forest, Gradient Boosting, ARIMA.
- Integrate real-time weather, demand, and supply data for robust forecasting.

Work Breakdown Structure





Key Deliverables

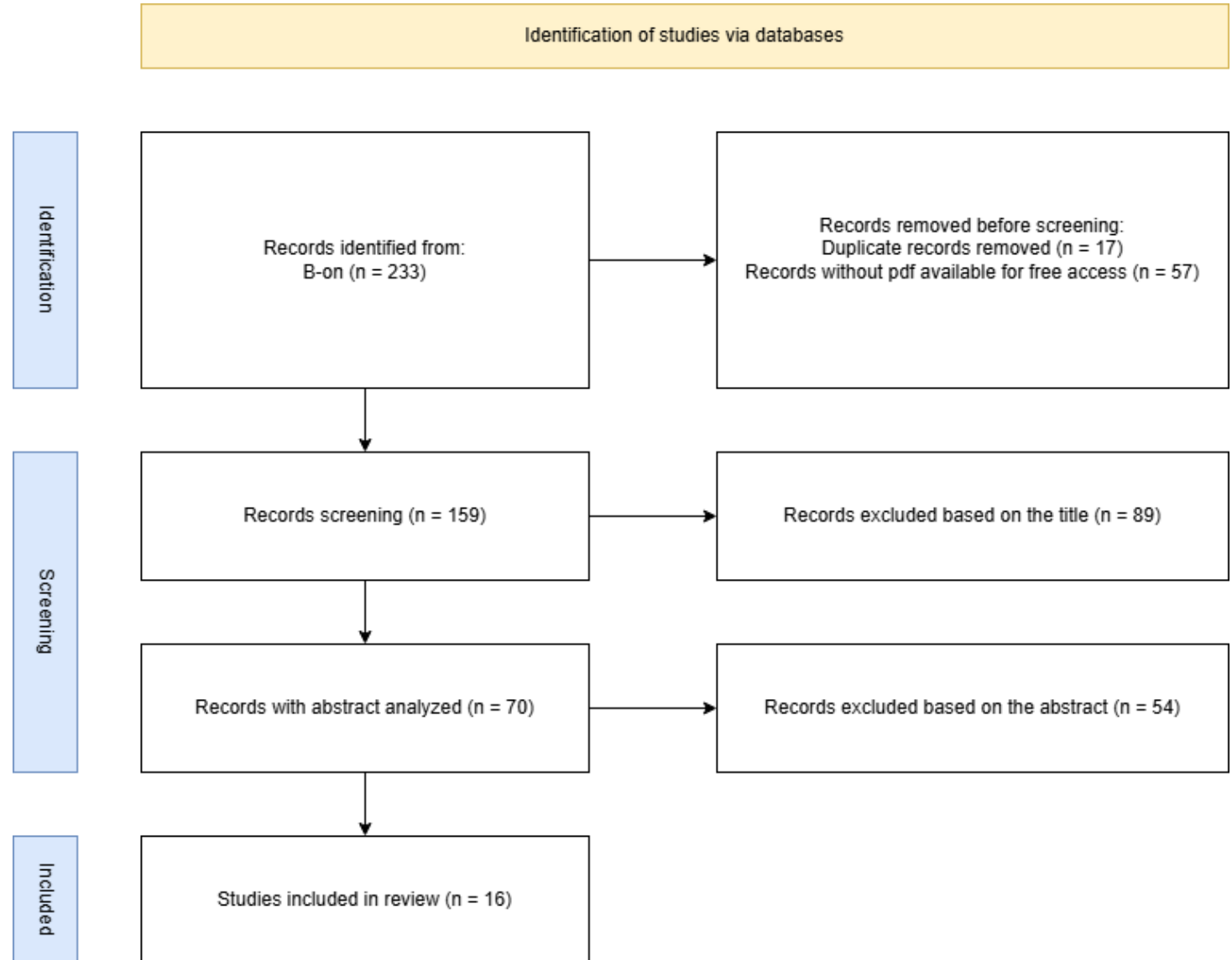
- Short-term and long-term forecasting models
- Presentation
- Dissertation

Research Questions

- How can machine learning techniques improve the accuracy of short-term electricity price forecasting in deregulated markets?
- Which machine learning algorithms provide the most accurate and robust forecasts for long-term electricity prices, considering factors such as economic trends and regulatory changes?
- What are the most relevant features that influence electricity price movements, and how can they be extracted from historical and real-time data?
- How can the integration of multiple data sources (e.g., weather data, demand forecasts, supply-side information) enhance the accuracy and reliability of the forecasting models?

Research Methodology - PRISMA

- **Database:** B-on
- **Keywords:**
 - Electricity price forecasting
 - Machine learning electricity market
 - Short-term electricity price prediction
 - Long-term electricity price forecasting
 - Deregulated electricity market
 - Deep learning
 - ARIMA, LSTM, Random Forest, Gradient Boosting forecasting



Key Findings and Insights

- **Enhanced Forecasting Accuracy**
 - Advanced machine learning methods outperform traditional models
 - Crucial for volatile energy markets where precise predictions matter
- **Complexity of Electricity Pricing**
 - Deregulated markets pose significant challenges
 - Requires adaptive models to address dynamic supply-demand
- **Importance of Data Quality**
 - High-quality data, especially for wind and solar power, is essential
 - Data accuracy directly impacts forecasting improvements
- **Methodological Rigor**
 - Diverse datasets and bias mitigation ensure fairness
 - Models are representative and inclusive of all market participants



Answer to the Research Questions

-
- **RQ1: How can machine learning techniques improve the accuracy of short-term electricity price forecasting in deregulated markets?**
 - Large datasets with historical prices, patterns of demand, and real-time data can be used to further improve the accuracy of short-term electricity price forecasting by machine learning techniques. Techniques such as LSTM, Random Forest, and Gradient Boosting allow capturing complex nonlinear relationships and interactions among several influencing factors that may be ignored by traditional models such as ARIMA. These models can then learn possibly through advanced algorithms-the shifting supply and demand, thus making better predictions in dynamic market conditions

Answer to the Research Questions

- **RQ2: Which machine learning algorithms provide the most accurate and robust forecasts for long-term electricity prices, considering factors such as economic trends and regulatory changes?**
 - The dissertation assesses several machine learning algorithms that could be used for the most accurate long-term forecast of electricity prices, namely, ARIMA, LSTM, Random Forest, and Gradient Boosting. Out of these, advanced methods include LSTM and Gradient Boosting, which are known for their capability to handle highly complicated datasets and capture long-run dependencies. These can, therefore, be ideal in price predictions that change due to economic trends and regulatory decisions. A comparison of such algorithms is made based on the accuracy, robustness, and computational efficiency of each.

Answer to the Research Questions

- **RQ3: What are the most relevant features that influence electricity price movements, and how can they be extracted from historical and real-time data?**
 - Relevant features that may impact the movement of electricity price include weather patterns, temperature, wind, solar demand trends, supply-side information about renewable energy availability, and economic indicators. These can be extracted from historical data by using different data preprocessing techniques such as handling missing values, outlier removal, and feature engineering. Real-time data can also be integrated for enhancing the feature set and enabling a more comprehensive study of factors affecting price fluctuations.

Answer to the Research Questions

-
- **RQ4: How can the integration of multiple data sources (e.g., weather data, demand forecasts, supply-side information) enhance the accuracy and reliability of the forecasting models?**
 - The integration of different data sources further enhances the models of forecast accuracy and reliability due to a holistic overview of various factors that can affect electricity prices. By incorporating, for example, weather data into demand forecasts and further information on supply, various interdependencies and correlations are more likely to be picked up by the model, which might not have been visible using one data source only. This comprehensive approach strengthens the models' adaptation capabilities in changing market conditions and enhances the robustness of the forecasts.



Methodology Overview

- Data Collection:
 - Historical electricity prices, weather data, demand patterns
- Model Development:
 - LSTM, Random Forest, Gradient Boosting, ...
- Evaluation Metrics:
 - MAE, RMSE, R^2
- Validation with real-world data: European Energy Exchange, NYISO



Contributions

- Enhanced models for electricity price forecasting.
- Practical tools for market participants (e.g., traders, grid operators).
- Promotes efficient energy trading and system stability.

Conclusion



Significant Improvement in Accuracy

Advanced machine learning techniques enhance electricity price forecasting accuracy, outperforming traditional models in volatile markets.



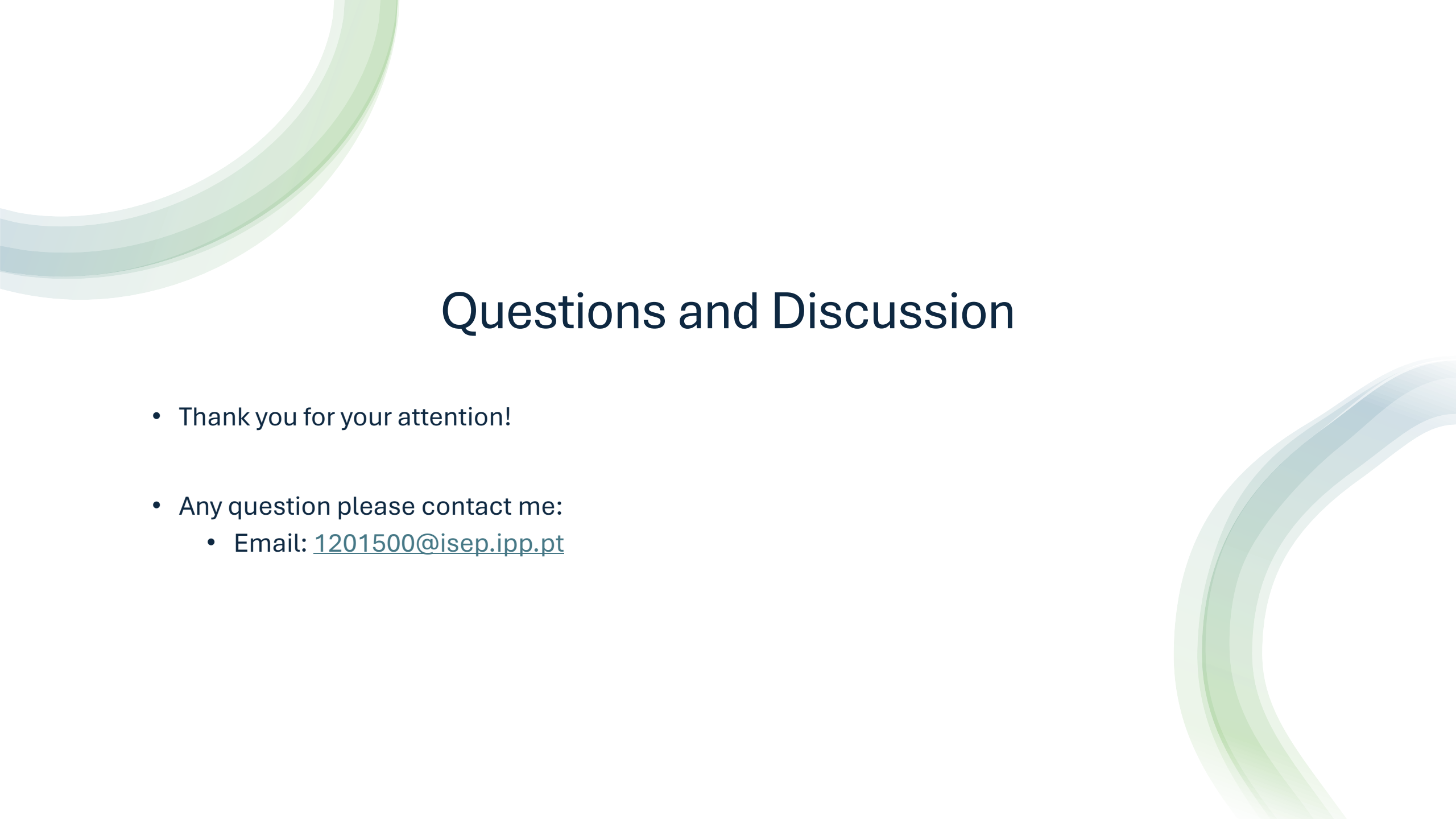
Complex Market Dynamics

The research highlights the complexities of deregulated electricity markets, necessitating sophisticated forecasting models that account for dynamic supply and demand factors.



Data Quality is Crucial

High-quality and real-time data, especially from renewable sources, is essential for developing reliable forecasting models.



Questions and Discussion

- Thank you for your attention!
- Any question please contact me:
 - Email: 1201500@isep.ipp.pt